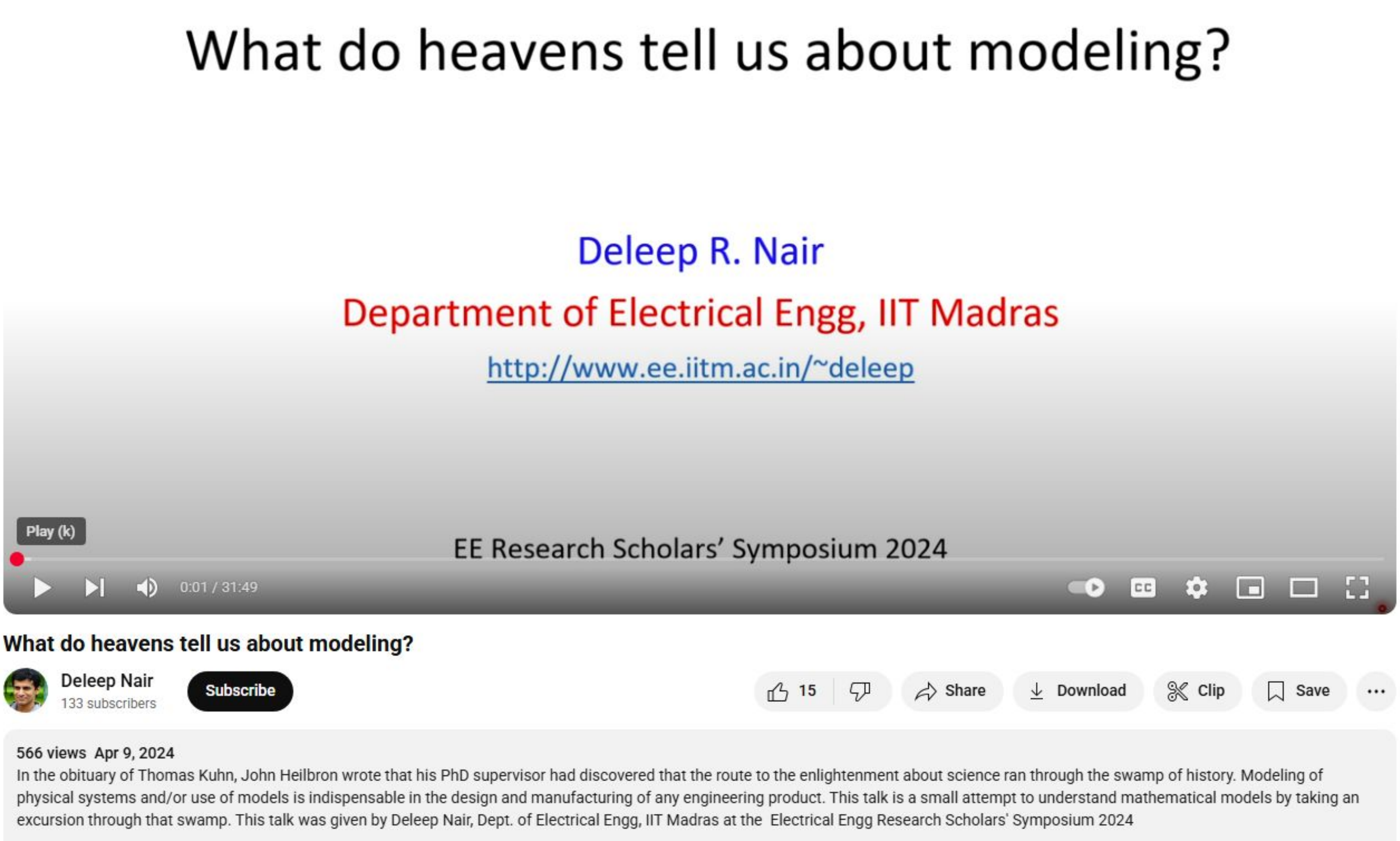
Compact Modeling

Deleep R. Nair

Modeling



https://www.youtube.com/watch?v=46Il1MvazF0

Types of Models

- Numerical, compact model, analytical, look up table based ones (LUT).
- Analytical – Physical understanding based on few parameters, closed form solution
- Numerical – Detailed understanding of device
- Compact – Needed for circuit simulation (eg: mobility)
- LUT – look up table is created based on device simulation and interpolations used

Compact model

- Particularly aimed for large scale circuit simulation
- Modern day microprocessor contains > 1 billion FETs
- Need to get meaningful results in few days
- Accuracy, generality and computational efficient
- Modern model consists of 300-400 equations and 150-400 parameters
- Digital to analog world – more restrictions on model equations, smoothing functions etc as they are worried about higher derivatives

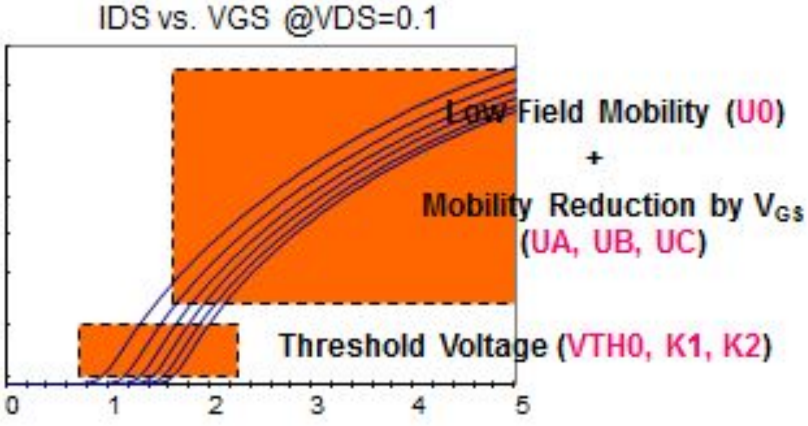
Models

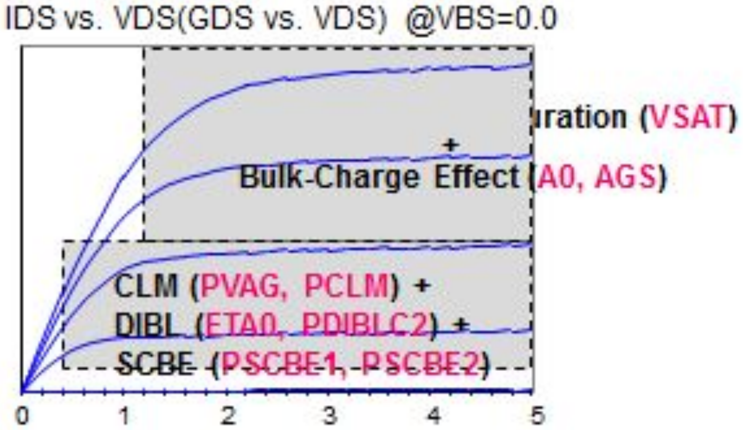
- V_T based – BSIM3, BSIM4 (problems in sub 100nm)
- Q_{inv} based – EKV (problem under accumulation for RF applns)
- Ψ_s based – HiSIM, PSP (selected by compact model council as follower to BSIM)

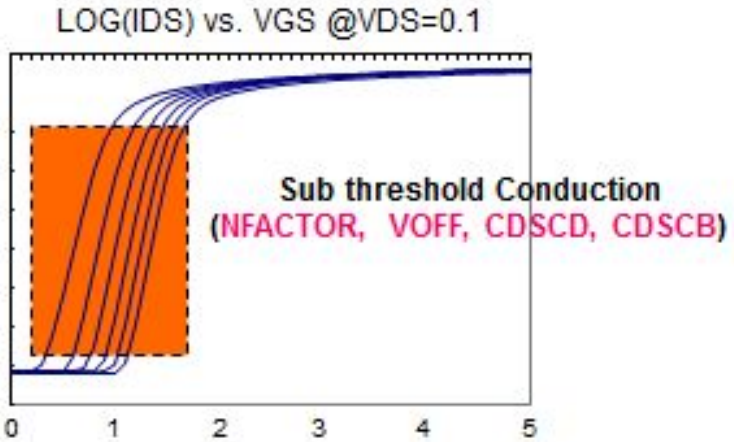
Modeling process flow

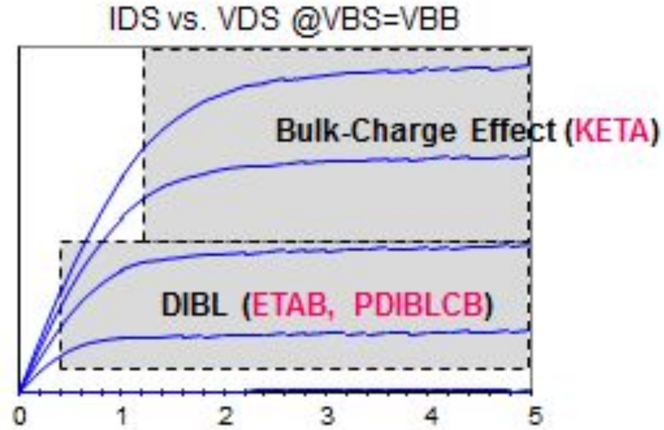
- Hardware Selection (best one available at that time – performance, electrostatics, centering)
- Data Collection (W,L, bias, Temp)
- Parameter Extraction/Fitting
- Hardware Fit Verification (Error % in Id-Vd/Vg, Vt vs L and W, gm – Vg, gds –Vd , C-V and other FET effects)
- Model Centering (Move the model to electrical targets)
- Physical Sanity Checks (Geometry, bias, temp and device consistency, corner check)
- Mathematical Sanity Checks (Negative derivatives, convergence test)
- Model Performance Checks (testing Ring Oscillator, Vdd and Temp, comparison to prev. version release)
- Release Testing

Parameter fitting (BSIM4)









Model evolution during tech development

Eval Model

- Formal performance commitment to design teams
- Match targets supplied by device designers
- Detailed IV curves patterned on previous hardware model
- Built before process is mature for centered hardware to be Available

Alpha Model

- Performance commitment doesn't change or may be uplifted
- As-fit models are based on hardware
- However, hardware does not meet device performance specifications
- As-fit models are adjusted or "centered" to meet technology performance commitments

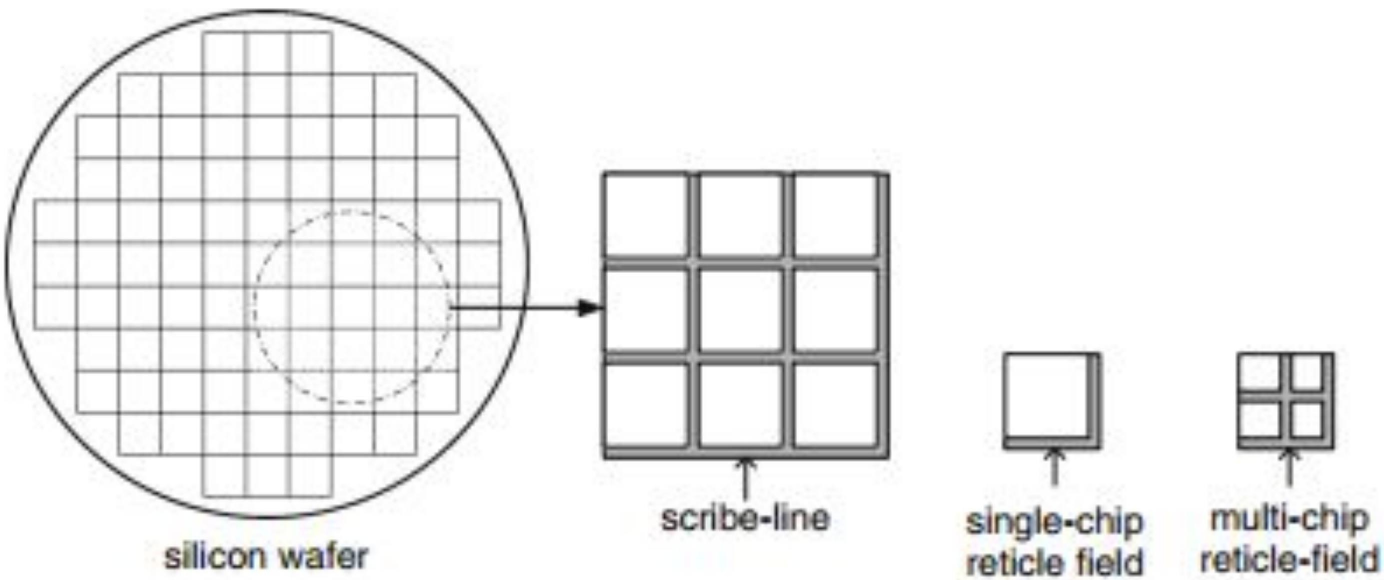
Beta Model

- Performance commitment doesn't change or may be uplifted
- Match centered hardware
- Based on "golden chip" with small adjustment to match line nominal targets

Why do you need test structures ?

- Testing and Verification – integral part of science and technology
- Indispensable tool for technology development
- Physical (GOX, Poly, Spacer, silicide, stress liner) or Electrical characterization
- Test structure content depends on
  - Development
  - Manufacturing

Wafer vs die (reticle) vs chip



Technology development phase

- Entire reticle field/die consist of test structures
- Optimization of process – to meet performance/leakage metrics
- Compact model development (alpha, beta ..)
- In line process monitoring (process can drift)
- Qualification of technology (reliability)
- More time devoted for testing

Manufacturing/Ramp up phase

- Product share space with test structures
- Design should be as close as one (dominant) used in product
- Limited test time and data
- Health of the line monitoring
- Sort chips (good vs catastrophic fails)

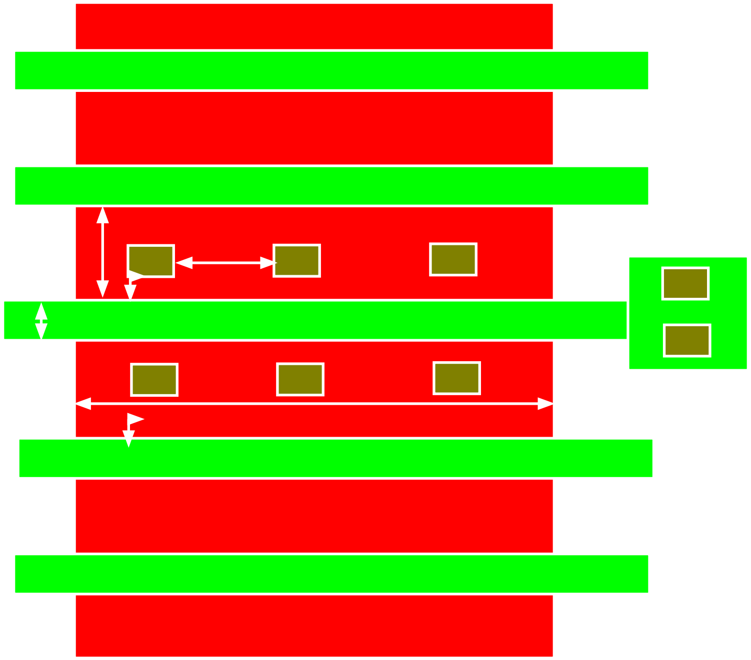
Test structure classification

- Function (silicon fabrication process, yield, model build, product debug and reliability)
- Circuit elements/ blocks under test (resistors, capacitors, MOSFETs, ring oscillators, logic circuits, and memory arrays)
- Placement on silicon (scribe line, test vehicle, on-product)
- Test metal layer (first, second, fourth, and last)
- Test type (DC parametric, low frequency (<10 MHz), high frequency and functional)
- Test equipment location (manufacturing in-line and laboratory/bench)

What we want to model ?

- I_d as f(V_{gs},V_{ds},V_{bs}, W, L, Temp)
- Capacitance (C_{gc}, C_{ov}, C_j, C_{jSTI}, C_{jswg})
- Contact/parasitic resistances
- Layout dependent performance
- Proximity effects
- Corner rounding
- Noise
- Mismatch
- Passive elements (resistors, capacitors, diodes etc)

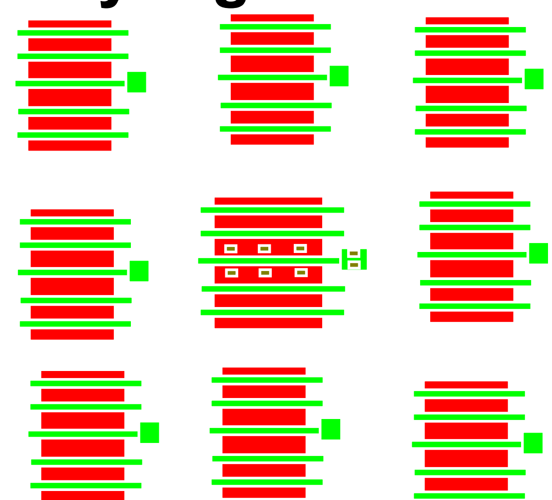
I_d as $f(V_{gs}, V_{ds}, V_{bs}, W, L, Temp)$



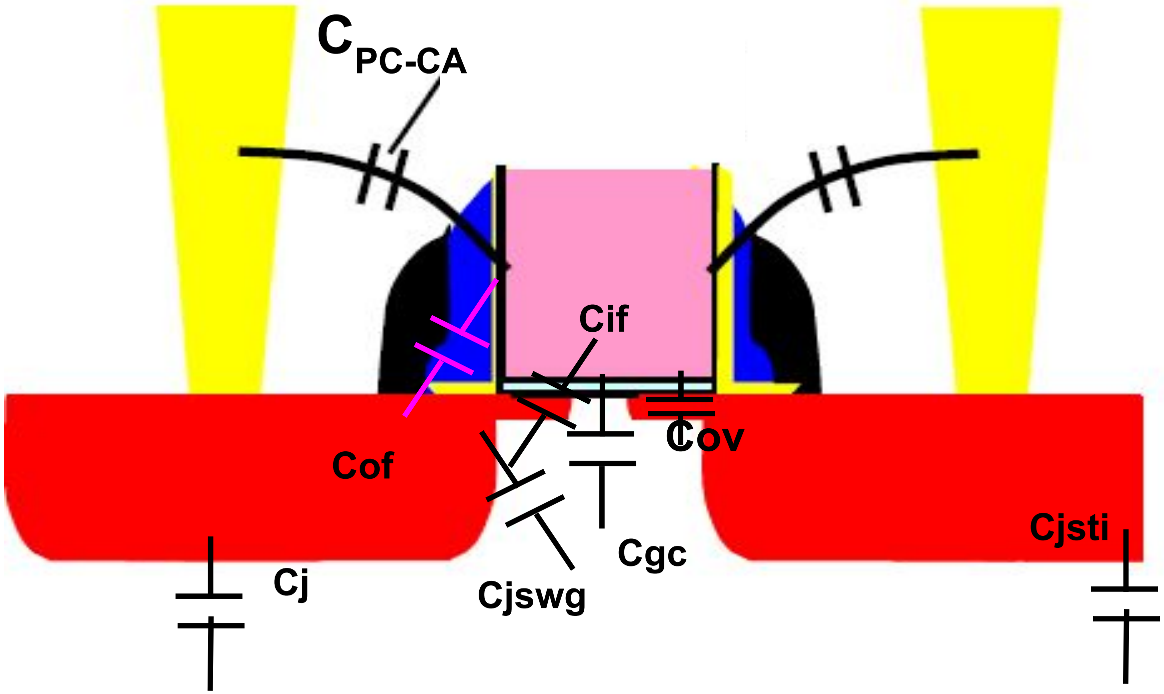
Ground rule clean structures
Different W, L combinations

Needed for different oxide and V_T flavors

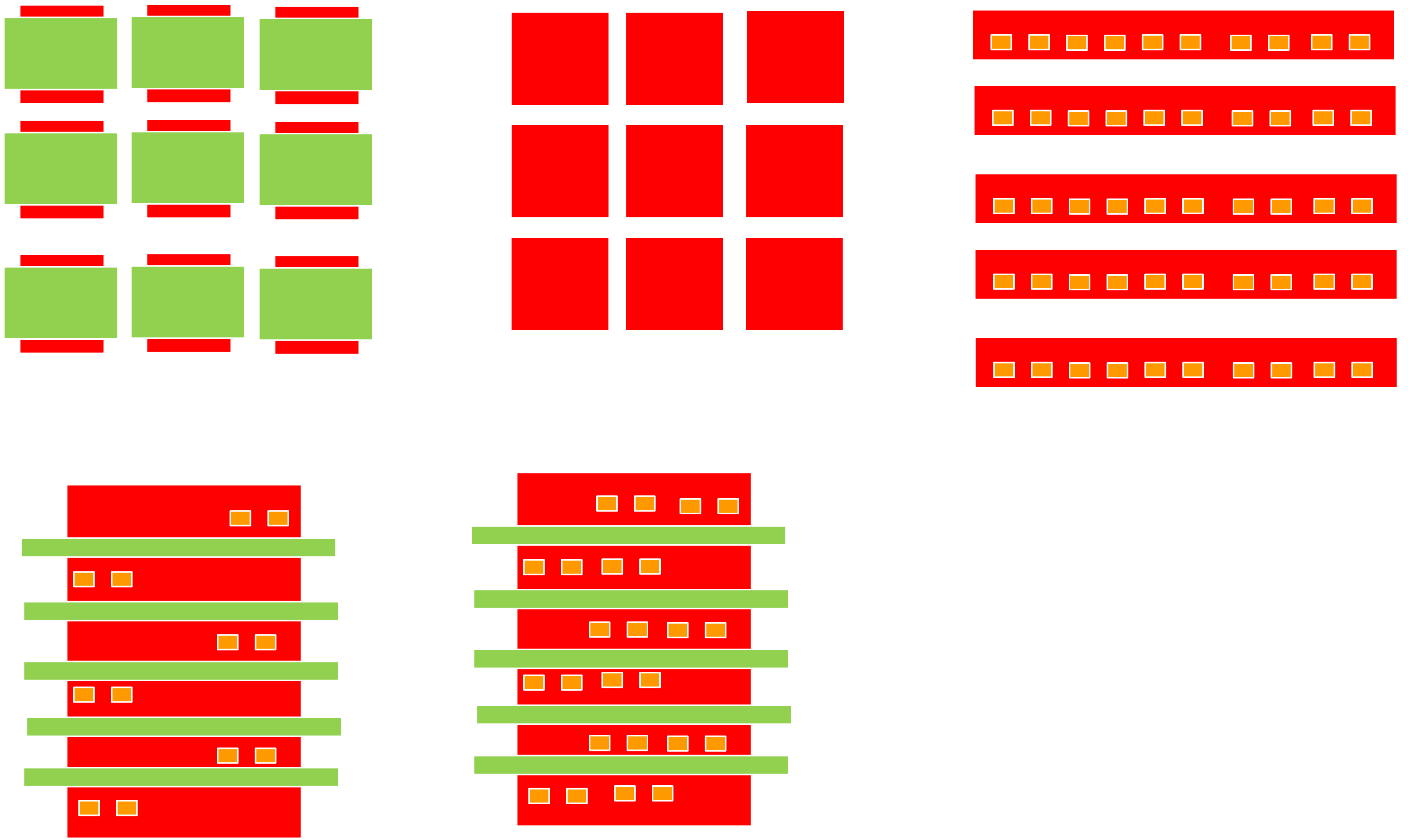
Placement of lot of dummy patterns around DUT to make field optically regular



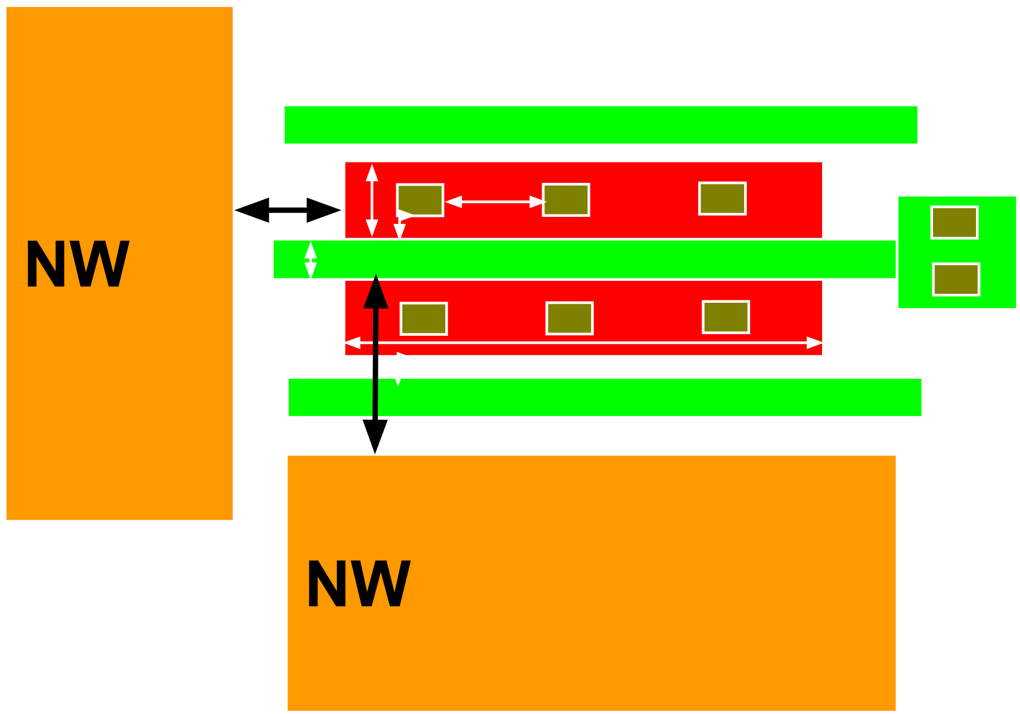
Capacitance (C_{gc}, C_{ov}, C_j, C_{jSTI}, C_{jswg})



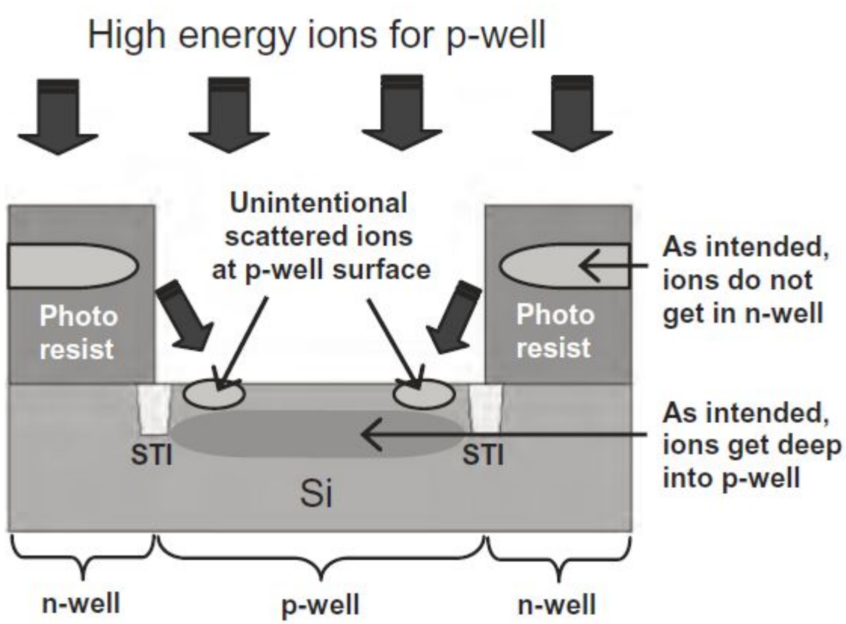
Capacitance (C_{gc}, C_{ov}, C_j, C_{jSTI}, C_{jswg})



Well proximity effect



- Tilted implants (2 deg)
- Resist edge may be tapered
- Ions scatter from edges and ends up in active area causing addn' V_T shift
- Higher shift for thick oxide devices



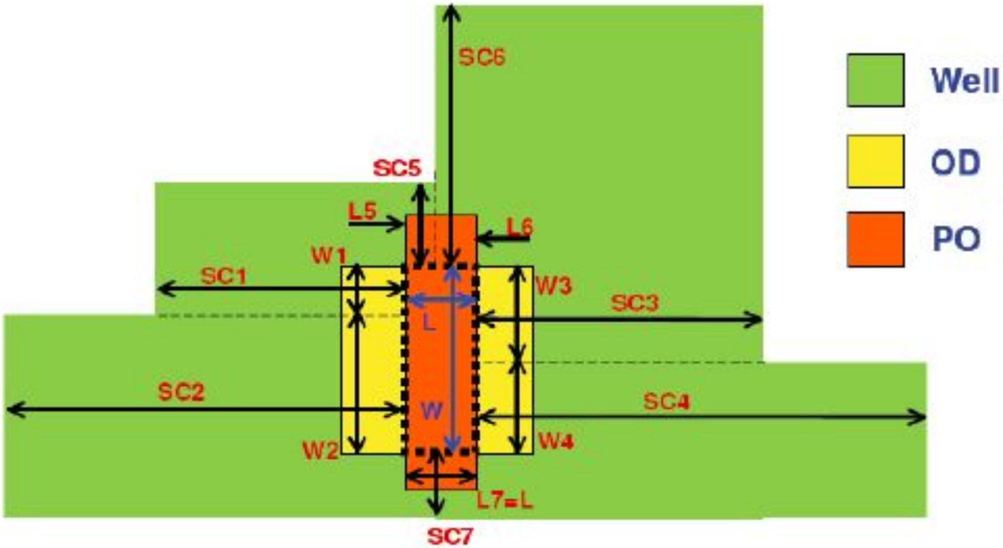
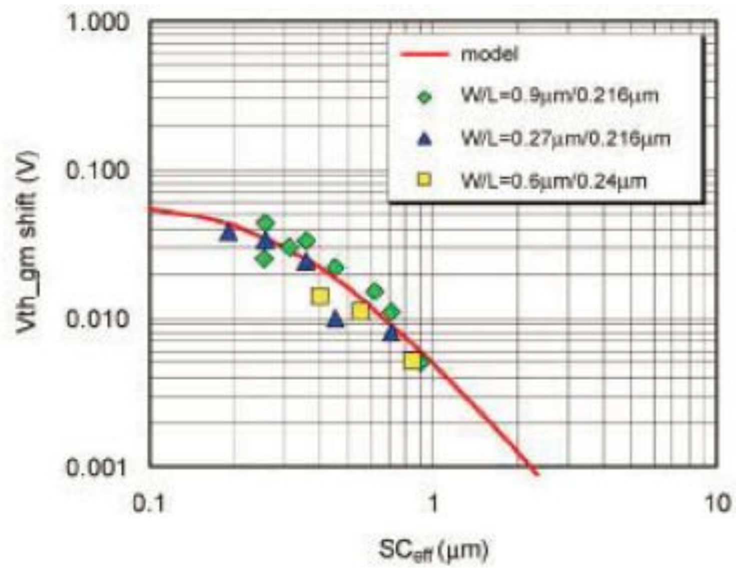
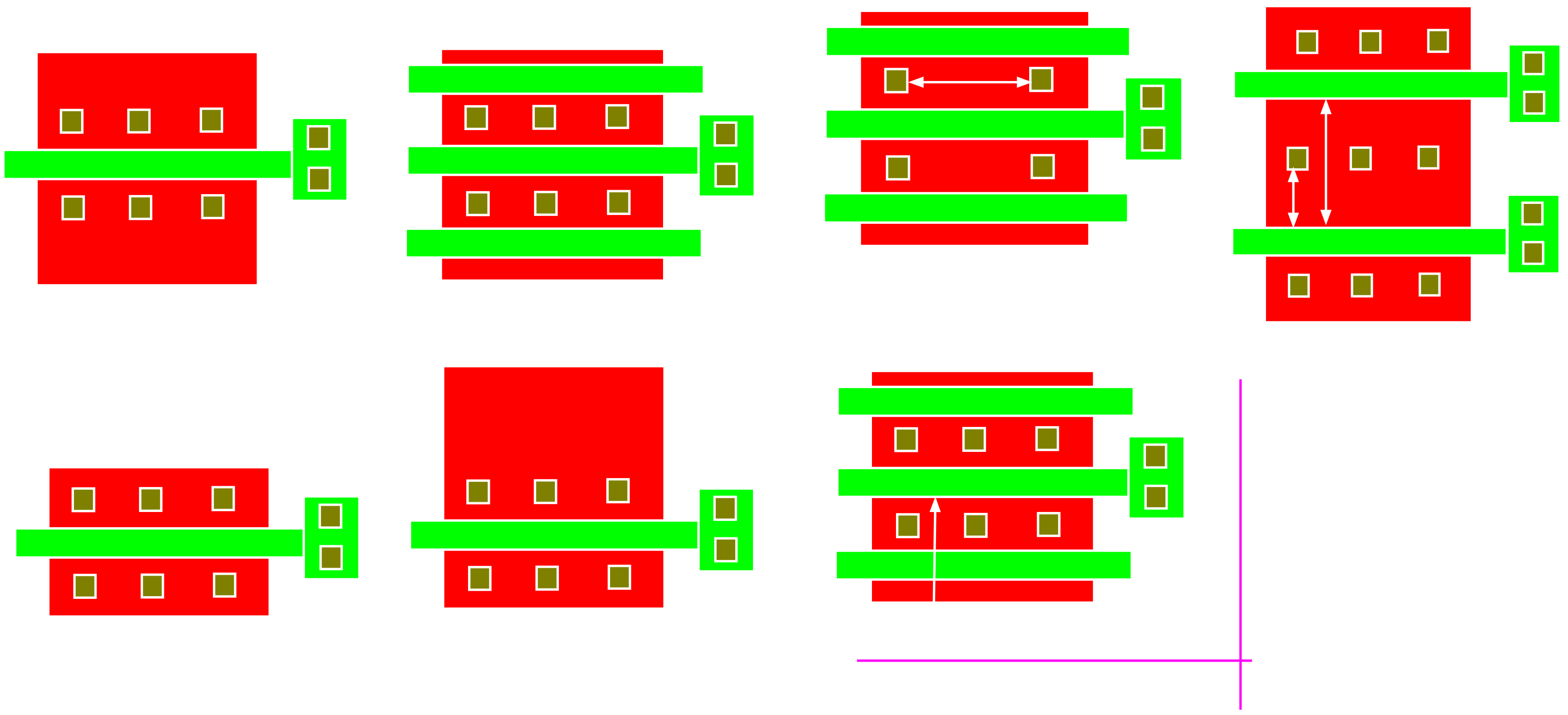


Fig. 3. WPE model layout measurements for the CMC WPE V2.2 model [2], [6]. OD is the oxide definition (active area). PO is the gate.



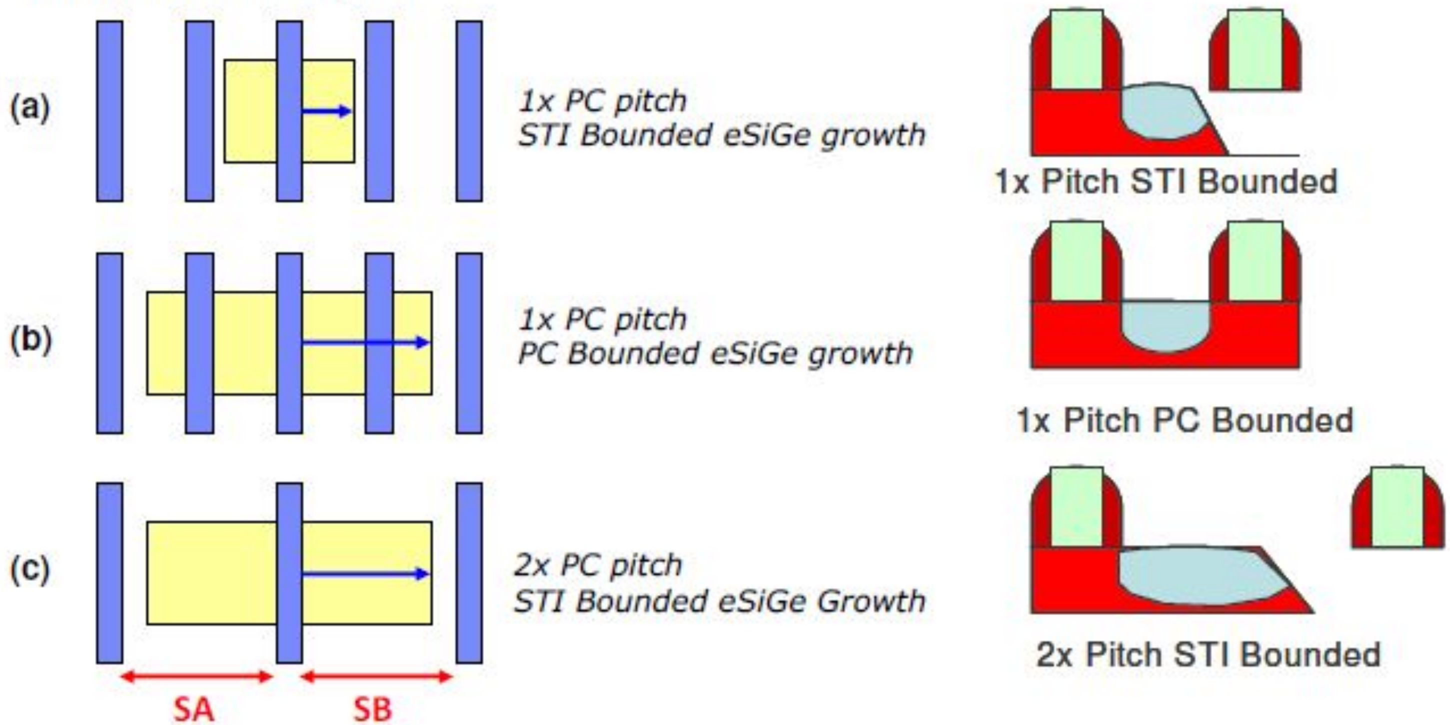
J. Faricelli. Layout-Dependent Proximity Effects in Deep Nanoscale CMOS **2010**

Stress impact



Stress introduced by liner films or epi S/D is proportional to volume and its proximity to channel

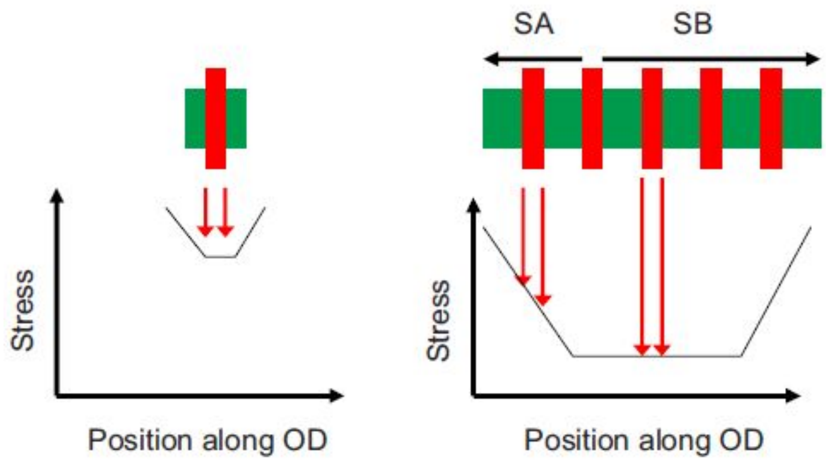
Example of eSiGe Layouts:



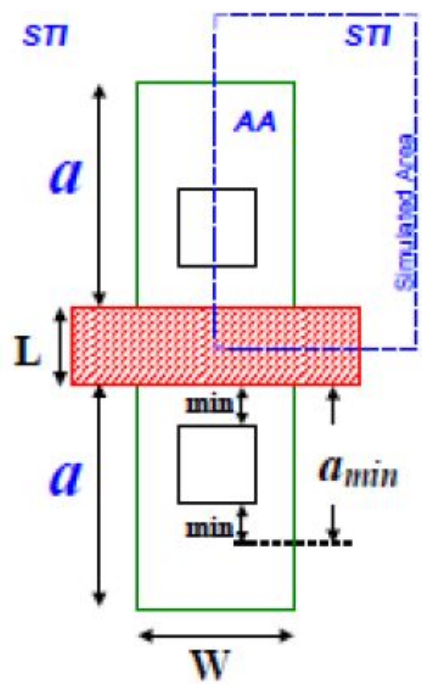
*SA/SB: Distance from active PC edge to active area edge from Source or Drain Side

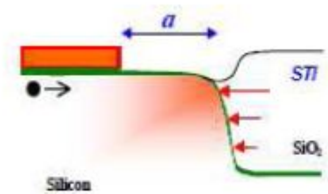
L. Song, VLSI-TSA 2012

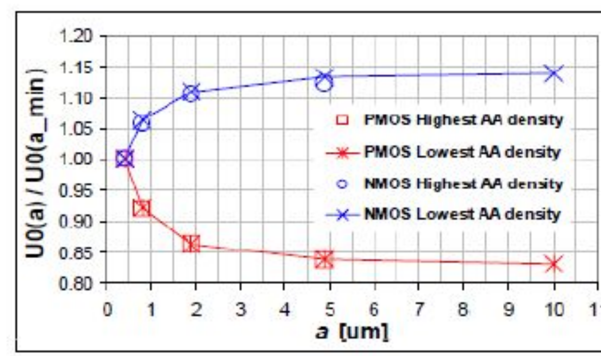
STI proximity



Compressive stress and reduction in Diffusivity (more recombination of Interstitials near STI edge, thereby reducing Transient Enhanced Diffusion)







RA Bianchi, IEDM 2002

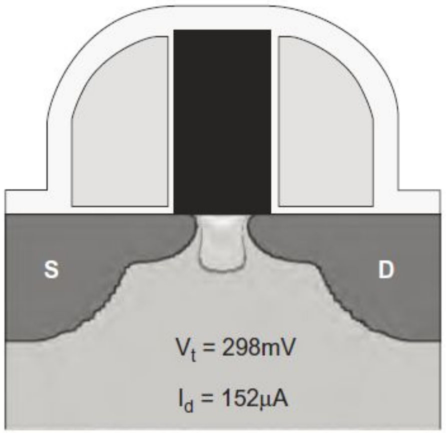
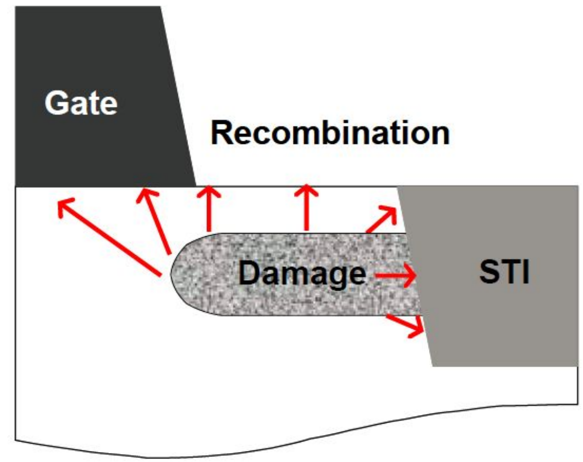


Figure 3.8 Nested nMOSFET with low V_t and strong I_d. Nested transistor with $L_s = L_d = 220$ nm is surrounded by similar transistors on the left and on the right, without STI nearby.



Moroz, ECS 2006

Tsuno, VLSI 2007

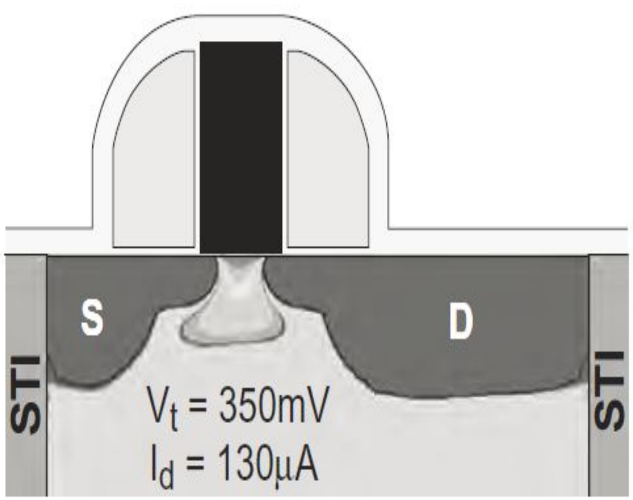


Figure 3.10 Isolated, asymmetric nMOSFET with small source and drain sizes, lea higher V_t and weaker I_d. Here, $L_s = 120$ nm and $L_d = 240$ nm.

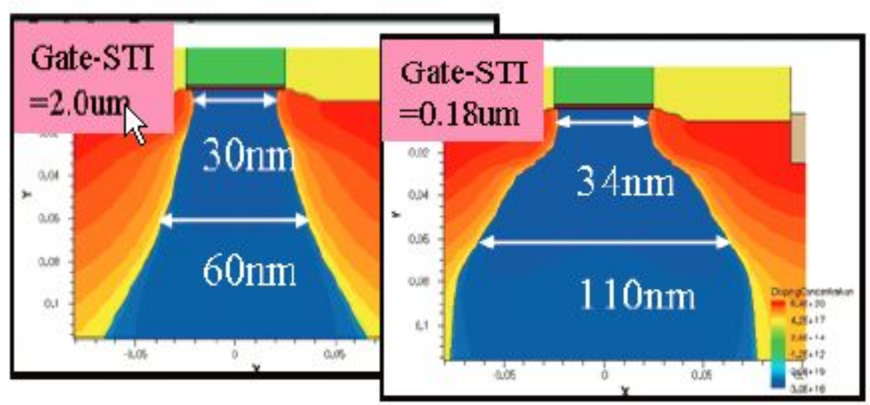
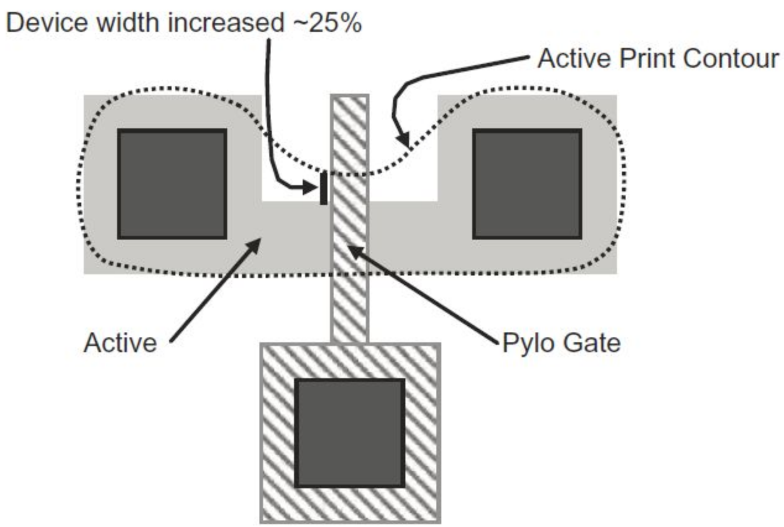
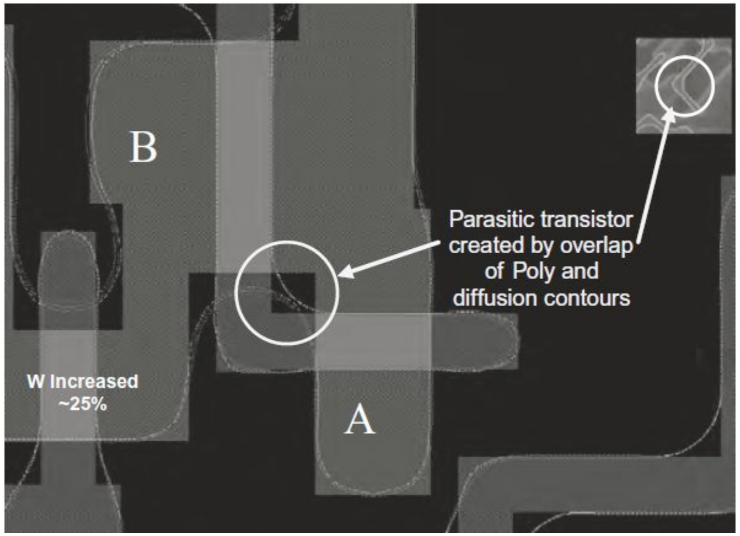
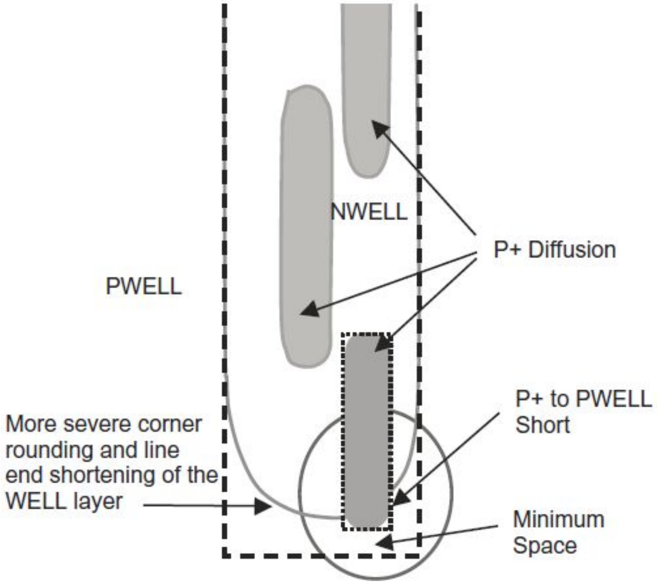


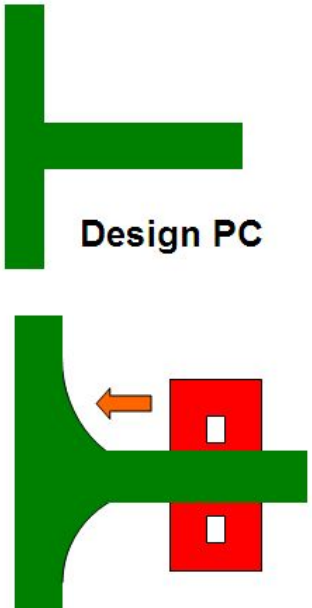
Fig. 4 Simulated difference of dopant distributior profile of an NFET as gate-STI shrinks. Both extension and S/D dopant diffusion are retarded.

Corner rounding



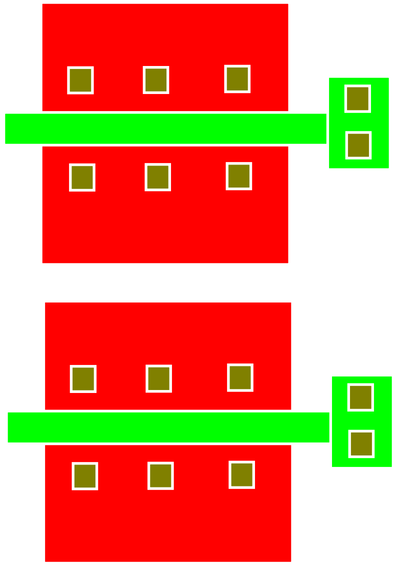






Wong et al: Nano CMOS design for DFM

Mismatch



Two transistors in close proximity
Similar layout dependent effects (same proximity)
V_T is measured on both

Take $\sigma(\Delta V_T)$. Plot it against $1/\sqrt{WL}$

$$\sigma V_{Tran} = \left(\frac{\sqrt[4]{4q^3\varepsilon_{si}\phi_B}}{2}\right)\cdot\frac{T_{ox}}{\varepsilon_{ox}}\cdot\left(\frac{\sqrt[4]{N}}{\sqrt{Weff\cdot Leff}}\right) =$$

Variability

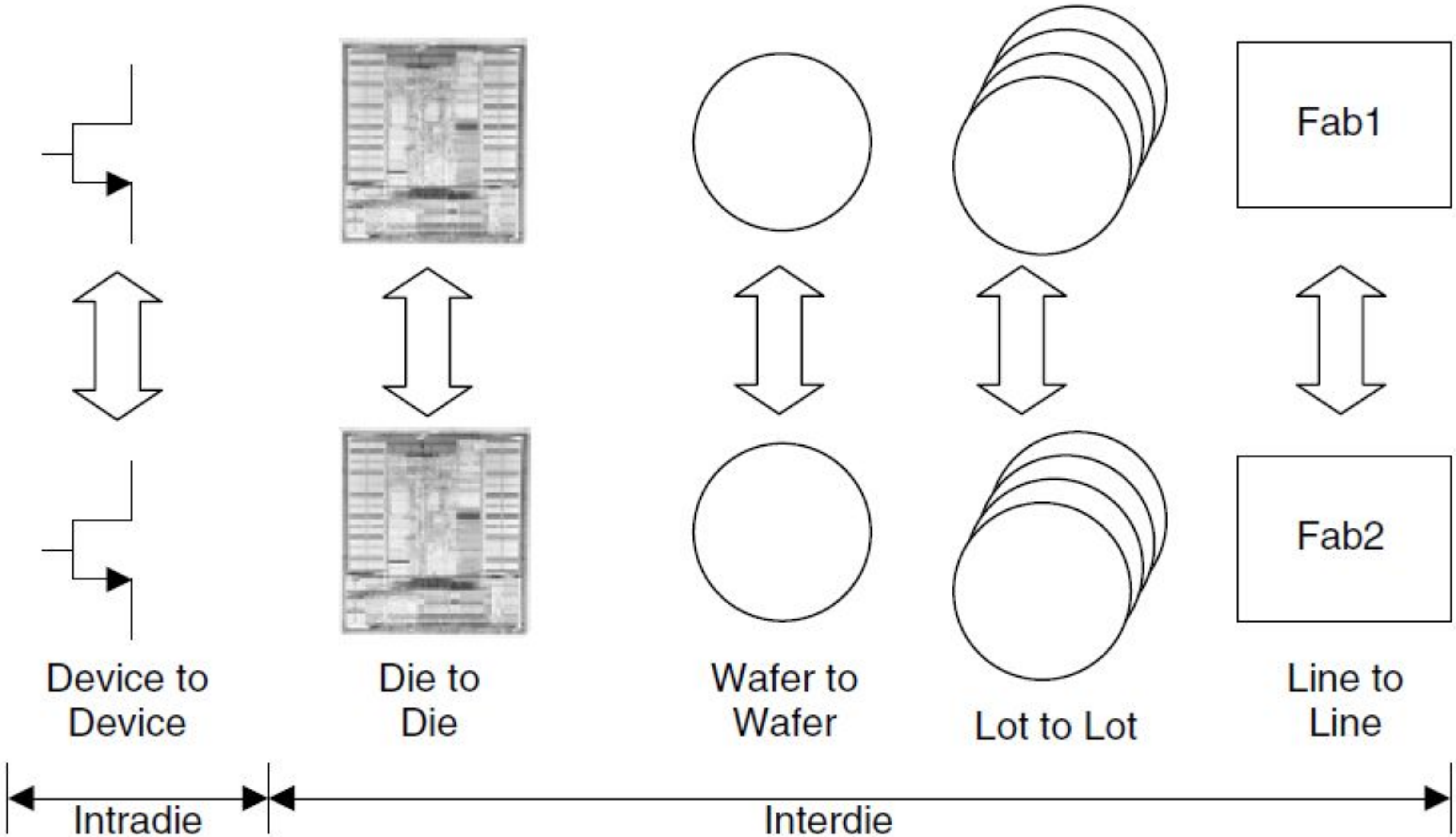
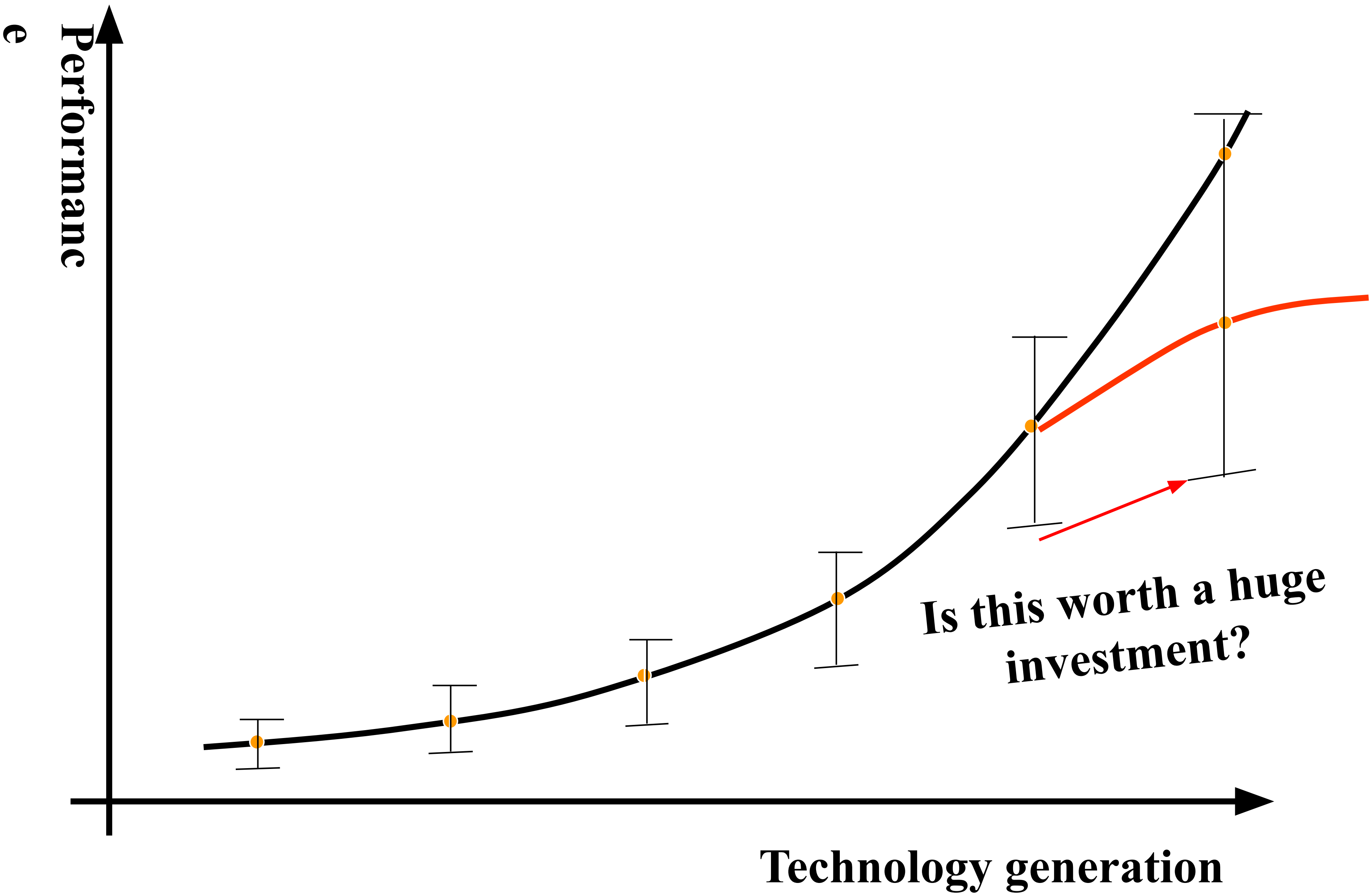


Figure 11.32 Various levels of process variations.

Variability eats into performance



Source of variations

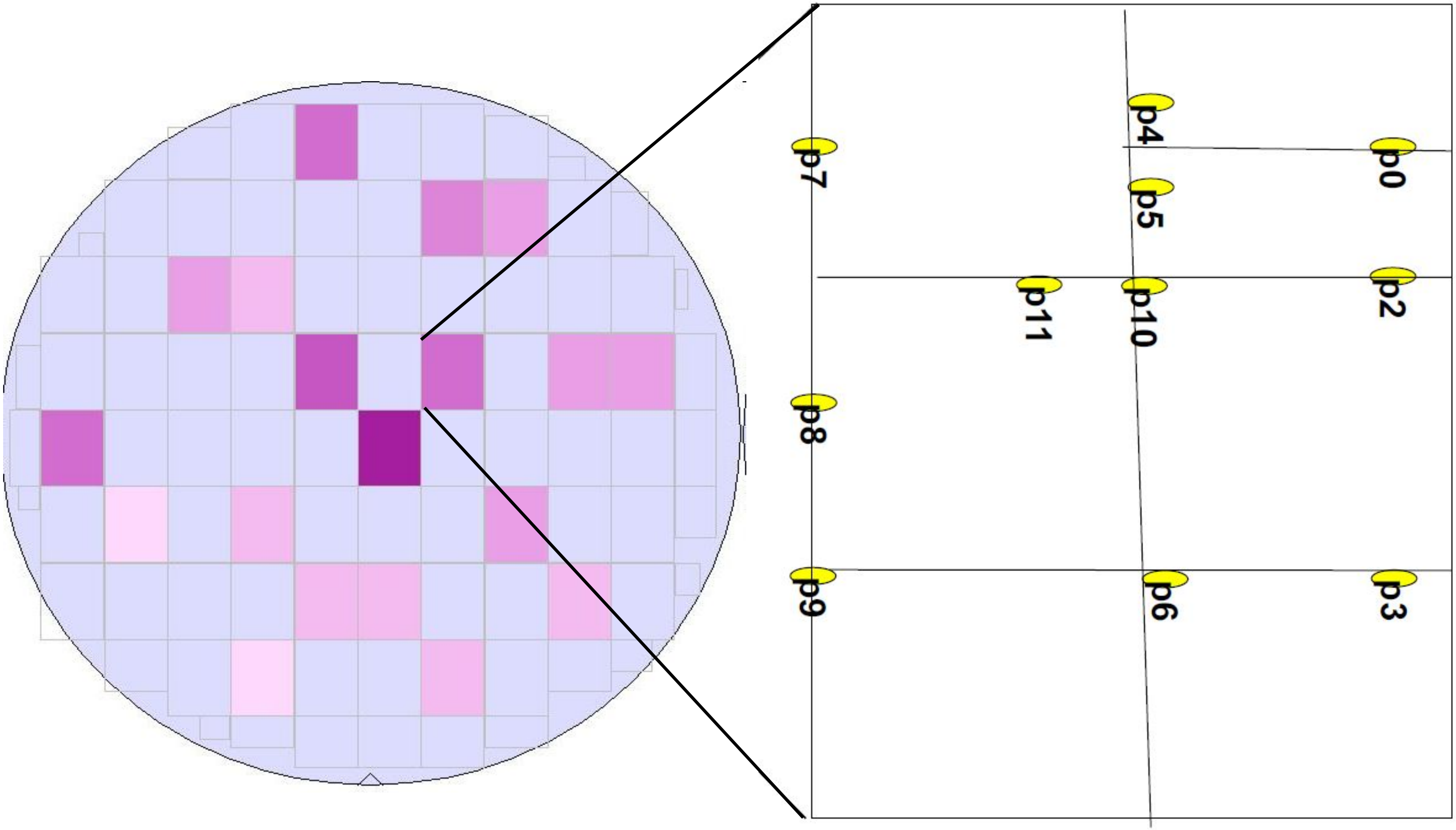
- Chip mean tolerance and Across chip variation
- Tinv/Lpoly/Wpoly
- Random dopant fluctuations
- Spcr loading (Poly density)
- RTA driven local temperature differences (Poly and active fill)
- CMP
- Work function (from HK/MG)
- Thickness of box/fin (ETSOI, FINFET)
- Systematic variation (Overlay, SRAM L-R mismatch)

Type of variations

Systematic variations can be predicted prior to fabrication; (layout dependent channel length variation). Successful technology scaling relies on the effective compensation of systematic variation components in both the process and design phases.

R*andom variations* are due to the inherent unpredictability of the semiconductor technology itself. (RDF, GOX thickness, interconnect thickness, dielectric permittivity)

Correlated vs uncorrelated variations (Vth =f(Vdd, Leff)



Chip mean tolerance vs Across chip variation

Chip mean tolerance (CMT)- Process varies from chip to chip & lot to lot causing random variation in behavior between chips of identical design

Across chip variation (ACV)- Process varies from device to device on a single chip causing random variation between devices and circuits of identical design

Methods to simulate variations

Fixed Corner methodology: Quick but sometimes be pessimistic or optimistic. Available as switch in netlist (0/1/2/..)

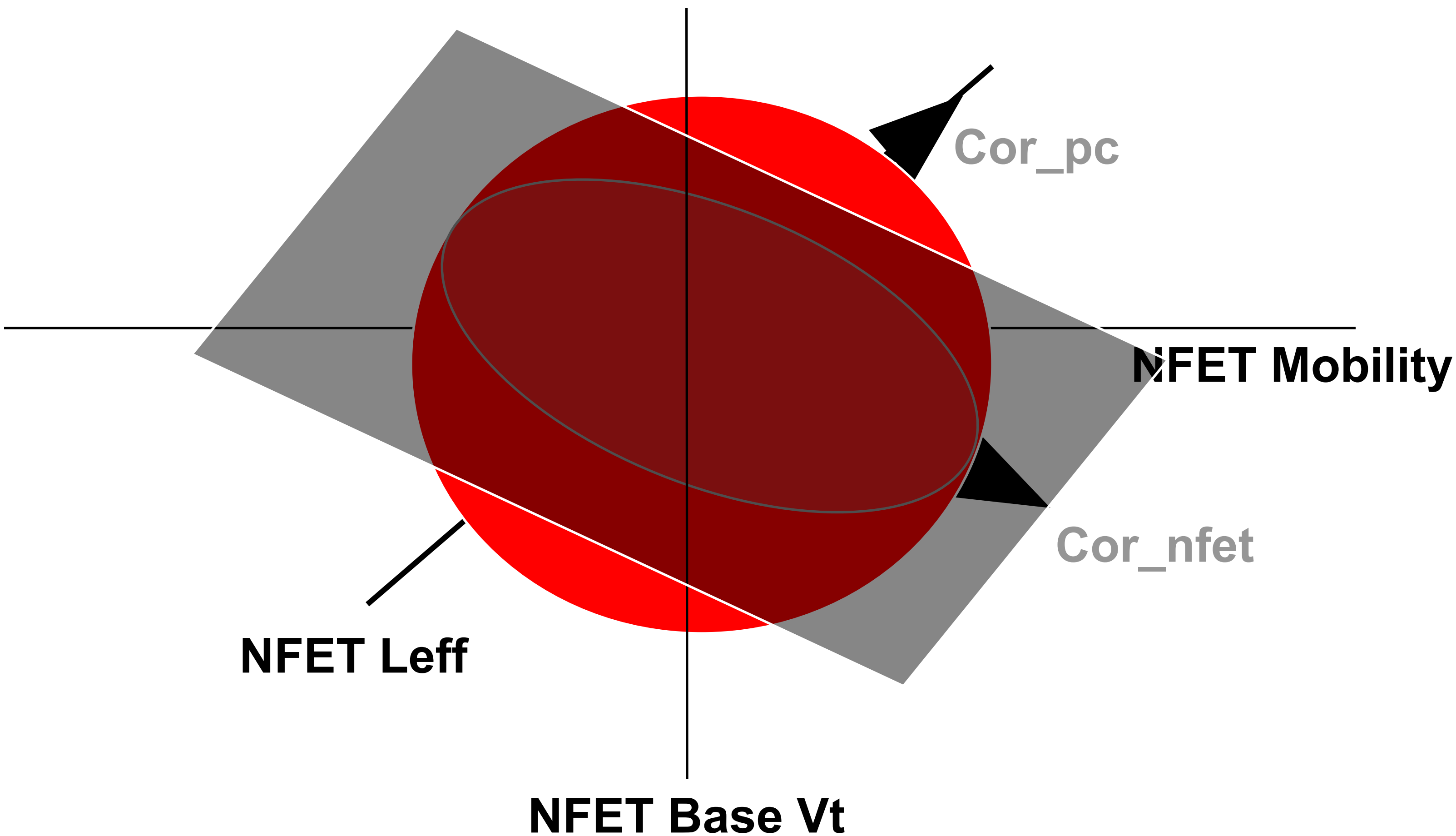
Monte Carlo simulation: Detailed and time consuming. Random variation of selected BSIM parameters. Depends on the accuracy of variations of model parameters

The Basis of the Process Model is Monte Carlo

- The Monte Carlo model specifies probability densities in high dimensional space

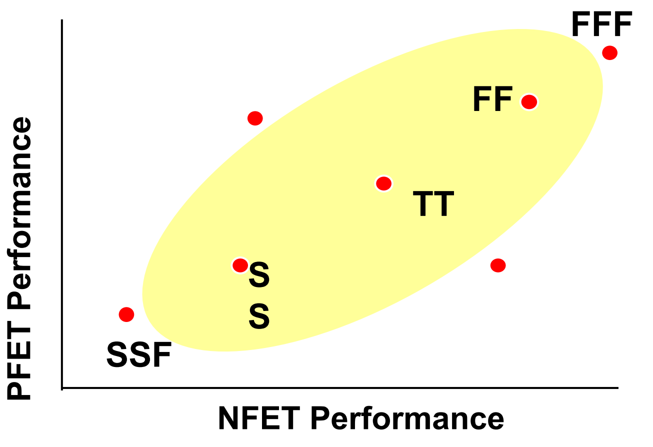
- User defined corners collapse that space onto a smaller number of dimensions

High Dimension MC to Lower Dimension Corner



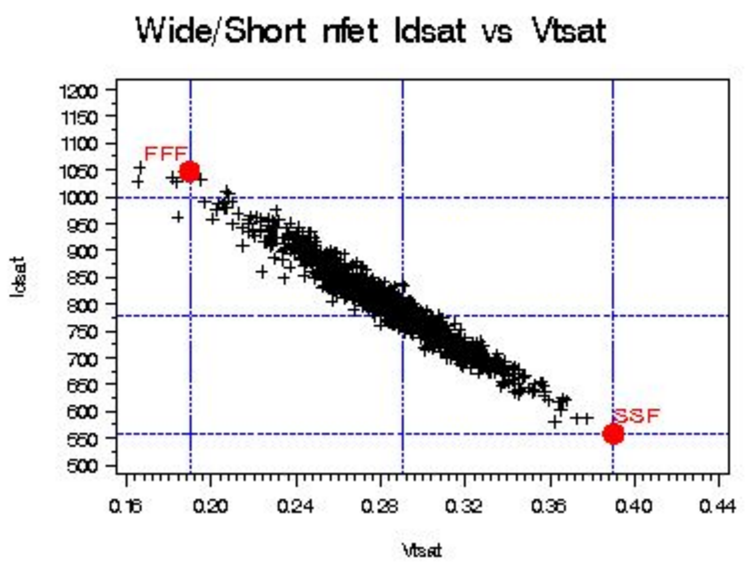
Fixed Corners

- **Fixed Corners are used as a low cost alternative to Monte Carlo simulations in circuit design**
- **Corner models are derived from a combination of Design Manual specs, Monte Carlo simulations,
and other information on process variation.**
  - **Improving performance in the corner models (tightening corners) is ultimately dependent on improving key technology tolerances (ΔL, Idsat, Tox…)**
- **Typical Fixed Corners would be**
  - **All devices fast**
  - **All devices slow**
  - **N-Type fast, P-Type slow**
  - **N-Type slow, P-Type fast**
  - **Other special requirements**



FFF/SSF: Functional Corners

- FFF/SSF corners are defined to match Design Manual 3σ Idsat, Idlin, Vtsat, and Vtlin tolerances
  - A set of BSIM parameters are skewed from nominal to find best sensible and probable solution.
    - Parameters include: TOXE, XL, XW, VTH0, U0, RDSW, PVTH0
    - Solution confined to region bounded by MC distribution 3σ limits and by requirement that all parameters must move at least some minimum amount.
    - TOXE, XL, XW move the same for all devices of similar Tox
    - Parameters must move in logical direction (i.e. U0 must be larger for FFF and slower for SSF)
    - Solutions are not required to be symmetric. Parameter delta from nominal for SSF does not have to be -1 times parameter delta for FFF.



FF/SS: Performance (Timing) Corners

- Corners set to match 3σ fast/slow ring oscillator delay
  - 1000 Monte Carlo runs made on ROs
  - NFET/PFET pair skewed equally in the direction of the FFF/SSF corner until corner delay matches Monte Carlo 3σ fast/slow limit.